Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

CompTIA Certified A+ Training (Core 1 and Core 2)

TGS Ref No: TGS-2024048317

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 7.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
6.0	1 March 2025	Previous release — 5-day lesson plan covering the nine A+ Core 1 and Core 2 topics.	Dr Alfred Ang
7.0	19 August 2026	Major revision — rebuilt against the current CompTIA A+ Core 1 (220-1101) and Core 2 (220-1102) exam domain weightings. Schedule now covers 54 hands-on labs across the five days, using the browser-based lab toolkit (IP Calculator, PCAP Analyzer, Cybersecurity Simulator, RegexLab) and the Killercoda Ubuntu playground.	Dr Alfred Ang

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Course Information

Course Title	CompTIA Certified A+ Training (Core 1 and Core 2)
WSQ Course Reference	TGS-2024048317
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
TSC Alignment	Infrastructure Support (ICT-OUS-3007-1.1)
Duration	5 days · 8 training hours per day (40 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks counted within training time)
Mode	Instructor-led, hands-on IT support labs using physical hardware where available and browser-based tools throughout
Certification Alignment	CompTIA A+ Core 1 (220-1101) and Core 2 (220-1102)
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Diagnose technical issues in network operations and implement procedures to resolve root causes.
- LO2: Troubleshoot technical issues and perform advanced infrastructure configurations.
- LO3: Develop action plans for upgrades and propose improvement ideas based on user needs.
- LO4: Test infrastructure systems and organise information for developing user guides.

Skills Framework Alignment

This course is aligned to the Skills Framework Technical Skill and Competency Infrastructure Support (ICT-OUS-3007-1.1).

Knowledge:

- K1 Diagnostic tools and processes to identify technical issues or disruptions in network infrastructure
- K2 Infrastructure and network configuration techniques
- K3 Troubleshooting techniques for infrastructure technical issues and problems
- K4 Potential benefits and impact of infrastructure upgrades and improvements
- K5 Documentation and user-guide development practices for infrastructure systems

Abilities:

- A1 Diagnose technical issues in network operations
- A2 Implement procedures to resolve root causes of technical issues
- A3 Perform advanced infrastructure configurations
- A4 Develop action plans for infrastructure upgrades

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- A5 Test infrastructure systems against operating requirements
- A6 Organise information for the development of user guides

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on support, configuration and troubleshooting tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 5 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Lab Toolkit

All hands-on labs use free, browser-based tools that require no local installation, so every learner works in an identical environment:

Tool	Link	Used for
IP Calculator	https://alfredang.github.io/ipcalculator/	Browser subnet calculator for IPv4 and IPv6 — CIDR, netmask, network and broadcast addresses, usable host ranges and batch processing with CSV export.
PCAP Analyzer	https://alfredang.github.io/pcapanalyzer/	Browser packet-capture analyser — protocol distribution, top talkers, top conversations and a packet table with per-packet detail. Files are parsed locally and never uploaded.
Cybersecurity Simulator	https://alfredang.github.io/cybersecuritysimulator/	Safe threat-simulation lab covering phishing, XSS, SQL injection, password strength, malware, ransomware, social engineering and data leakage.
RegexLab	https://alfredang.github.io/regexgenerator/	Live regular-expression tester with flags, match explanation and substitution — used to filter and parse support logs.
Killercoda Ubuntu Playground	https://killercoda.com/playgrounds/scenario/ubuntu	Free browser-based Ubuntu terminal with root access — no install required. Used for every Linux command-line lab in this course.

Course Schedule

Day 1 — Mobile Devices and Networking

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules,

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		learning outcomes, course outline and mandatory digital attendance (AM)
10:00–11:00	60 min	**Topic 1 — Mobile Devices: laptop hardware and field-replaceable units, display technologies (LCD/IPS/VA/OLED), digitizers, connection methods, docking (concepts + demo)**
11:00–11:15	15 min	Tea break
11:15–13:30	135 min	Hands-on: Lab 1: Laptop Teardown and Field-Replaceable Unit Identification; Lab 2: Mobile Display Technology and Digitizer Fault Diagnosis; Lab 3: Mobile Connectivity — Bluetooth, NFC, Hotspot and Cellular Configuration; Lab 4: MDM Policy Design and Mobile Synchronisation
13:30–14:30	60 min	Lunch break
14:30–16:00	90 min	**Topic 2 — Networking: TCP/UDP, the A+ port list, network devices, wireless standards and bands, SOHO services (concepts + demo). Digital attendance (PM)**
16:00–16:15	15 min	Tea break
16:15–18:15	120 min	Hands-on: Lab 5: Protocols and Ports — Building the A+ Port Reference; Lab 6: IPv4 Subnetting and Address Planning with IP Calculator; Lab 7: Network Devices and Traffic Behaviour Analysis
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Hardware, Virtualization and Cloud

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Topic 2 continues — channel planning, SOHO configuration, cabling and packet analysis. Hands-on: Lab 8: Wireless Standards, Channel Planning and Interference; Lab 9: SOHO Network Configuration — DHCP, DNS, NAT and Port Forwarding
11:00–11:15	15 min	Tea break
11:15–13:30	135 min	Hands-on: Lab 10: Network

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		Cabling — Termination, Testing and Tool Selection; Lab 11: Packet Capture Analysis — Reading Real Network Traffic. Topic 3 — Hardware: cables and connectors, RAM, storage and RAID (concepts)
13:30–14:30	60 min	Lunch break
14:30–16:00	90 min	Hands-on: Lab 12: Cables and Connectors — Identification and Selection; Lab 13: RAM Identification, Installation and Channel Configuration; Lab 14: Storage Devices and RAID Level Selection. Digital attendance (PM)
16:00–16:15	15 min	Tea break
16:15–18:15	120 min	Topic 3 continues — motherboards, BIOS/UEFI, CPUs and cooling, power supplies, printers. Hands-on: Lab 15: Motherboard, BIOS/UEFI and CMOS Configuration; Lab 16: CPU Architecture, Sockets and Cooling Solutions
18:15–18:30	15 min	Day 2 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Hardware and Network Troubleshooting

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 17: Power Supply Sizing, Connectors and Safety; Lab 18: Printer Installation, Configuration and the Laser Imaging Process
11:00–11:15	15 min	Tea break
11:15–13:30	135 min	**Topic 4 — Virtualization and Cloud Computing: hypervisor types, VM resource planning, cloud service and deployment models (concepts). Hands-on: Lab 19: Hypervisor Types and Virtual Machine Resource Planning; Lab 20: Building and Configuring a Linux Virtual Machine; Lab 21: Cloud Service and Deployment Model Selection**
13:30–14:30	60 min	Lunch break
14:30–16:00	90 min	**Topic 5 — Hardware and Network Troubleshooting:**

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		the CompTIA six-step methodology, POST and boot faults (concepts). Hands-on: Lab 22: Applying the CompTIA Six-Step Troubleshooting Methodology; Lab 23: POST, Boot and Power Fault Diagnosis. Digital attendance (PM)
16:00–16:15	15 min	Tea break
16:15–18:15	120 min	Hands-on: Lab 24: Storage and RAID Fault Diagnosis with S.M.A.R.T.; Lab 25: Display and Projector Fault Diagnosis; Lab 26: Mobile Device Hardware Fault Diagnosis
18:15–18:30	15 min	Day 3 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 4 — Operating Systems and Security

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 27: Printer Fault Diagnosis from Print Defects; Lab 28: Network Fault Diagnosis — Command Line to Packet Level. End of Core 1 domains
11:00–11:15	15 min	Tea break
11:15–13:30	135 min	**Topic 6 — Operating Systems: Windows editions and installation, partitioning, command line, Windows tools (concepts). Hands-on: Lab 29: Windows Editions, Installation Types and Partitioning; Lab 30: Windows Command Line — Navigation, Files and Copy Operations; Lab 31: Windows Networking and Repair Commands**
13:30–14:30	60 min	Lunch break
14:30–16:00	90 min	Hands-on: Lab 32: Windows Management Tools — Task Manager, MMC and Snap-ins; Lab 33: File Systems, Permissions and Share Configuration; Lab 34: Linux Command Line Essentials on Killercoda. Digital attendance (PM)
16:00–16:15	15 min	Tea break
16:15–18:15	120 min	Hands-on: Lab 35: Linux

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		Permissions, Ownership and User Management; Lab 36: macOS Tools and Cross-Platform File System Compatibility. Topic 7 — Security: physical and logical controls, authentication (concepts). Hands-on: Lab 37: Physical and Logical Security Control Design
18:15–18:30	15 min	Day 4 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 5 — Software Troubleshooting, Operational Procedures and Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 4 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Topic 7 continues — malware, social engineering, wireless and workstation hardening. Hands-on: Lab 38: Password Strength Analysis and Authentication Policy; Lab 39: Phishing Recognition and Social Engineering Defence; Lab 40: Malware Types, Symptoms and the Seven-Step Removal Procedure
11:00–11:15	15 min	Tea break
11:15–12:45	90 min	Hands-on: Lab 41: Network Attack Recognition — Injection, XSS and Data Leakage; Lab 42: Wireless Security and SOHO Router Hardening; Lab 43: Windows Security Configuration and BitLocker Encryption; Lab 44: Data Destruction, Disposal and Regulated Data Handling
12:45–13:30	45 min	**Topic 8 — Software Troubleshooting: Windows symptoms and recovery tools, malware and browser symptoms (concepts). Hands-on: Lab 45: Windows OS Symptom Diagnosis and Recovery Tools**
13:30–14:30	60 min	Lunch break
14:30–15:45	75 min	Hands-on: Lab 46: Log Analysis with Regular Expressions; Lab 47: Malware Symptom Response and Browser Problem Resolution; Lab 48: Mobile OS and Application Troubleshooting; Lab 49: System Restore, Backup Verification and

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		Recovery Testing. Digital attendance (PM)
15:45–16:15	30 min	**Topic 9 — Operational Procedures: documentation, change and asset management, backup, safety, professionalism. Hands-on: Lab 50: Ticketing, Documentation and Knowledge Base Authoring; Lab 51: Change Management and Asset Management; Lab 52: Backup Strategy Design and the 3-2-1 Rule; Lab 53: Safety Procedures, ESD Control and Environmental Compliance; Lab 54: Professional Communication, Scripting and Remote Access**
16:15–16:30	15 min	**Course recap, exam preparation guidance, TRAQOM survey and Briefing for Assessment**
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
17:30–18:30	60 min	**Practical Performance (PP) — hands-on support, configuration and troubleshooting tasks, 1 hour, open book. Assessment digital attendance**

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to CompTIA A+ exam domains)

Topic / Exam domain	Exam	Exam weighting	Course time	Labs
Topic 01: Mobile Devices	Core 1	13% of Core 1	7%	Lab 1, Lab 2, Lab 3, Lab 4
Topic 02: Networking	Core 1	23% of Core 1	12%	Lab 5, Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11
Topic 03: Hardware	Core 1	25% of Core 1	13%	Lab 12, Lab 13, Lab 14, Lab 15, Lab 16, Lab 17, Lab 18
Topic 04: Virtualization and Cloud Computing	Core 1	11% of Core 1	6%	Lab 19, Lab 20, Lab 21
Topic 05: Hardware and Network Troubleshooting	Core 1	28% of Core 1	15%	Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27, Lab 28
Topic 06: Operating	Core 2	28% of Core 2	17%	Lab 29, Lab 30, Lab 31, Lab 32,

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Systems				Lab 33, Lab 34, Lab 35, Lab 36
Topic 07: Security	Core 2	28% of Core 2	17%	Lab 37, Lab 38, Lab 39, Lab 40, Lab 41, Lab 42, Lab 43, Lab 44
Topic 08: Software Troubleshooting	Core 2	23% of Core 2	8%	Lab 45, Lab 46, Lab 47, Lab 48, Lab 49
Topic 09: Operational Procedures	Core 2	21% of Core 2	5%	Lab 50, Lab 51, Lab 52, Lab 53, Lab 54

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